

The curvilinear relationship between team member boundary-spanning differentiation and creativity

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Summary

Team creativity, as one of the most important sources of organizational competitiveness and continuous innovation, requires rich and non-redundant information from both internal and external sources. Although an emerging line of research has shed light on the role of team boundary-spanning activity in promoting creativity, it has solely focused on the mean level of team boundary-spanning activity and rarely addressed the fact that not all team members engage in boundary-spanning behaviors to the same extent. In this study, we broaden the existing knowledge by examining *when* and *how* the pattern of team boundary spanning, particularly the differentiation in team member boundary-spanning activities, promotes team creativity. Based on the categorization-elaboration model, we propose a curvilinear relationship between team member boundary-spanning differentiation and creativity that is mediated through information sharing. Further, we add an important layer of understanding to the nature of this curvilinear relationship by identifying team goal interdependence as an important moderator. Using multisource data collected from two time points in 70 teams, we have received general support for our hypotheses. As such, we enrich the conversation about the importance of team boundary spanning by answering “*Whether* team boundary-spanning differentiation fosters creativity, and *when* and *how*?”

KEYWORDS

team creativity, team goal interdependence, team information sharing, team member boundary-spanning differentiation (TMBSD)

1 | INTRODUCTION

Team creativity, as one of the most important sources of organizational competitiveness and continuous innovation, has led to the rapid growth of interest among the practitioners and academic communities (Hoever et al., 2012; Nouri et al., 2014). Although there is considerable evidence that a variety of antecedents can contribute to team creativity (e.g., team structure, composition, climate, and leadership style; Anderson et al., 2014), there is also a consensus that the promotion of team creativity usually requires rich and non-redundant informational inputs from members (Gong et al., 2013). Given the growing recognition that team boundary-spanning activity is crucial in

connecting teams with various internal and external sources and establishing important channels to acquire new knowledge and information, an emerging line of research has focused on the role of team boundary-spanning activity in supplementing the existing knowledge repository and generating novel ideas (e.g., Somech & Khalaili, 2014).

Although research over the past two decades has indicated that team boundary-spanning activity is inextricably linked to team functioning (Ancona & Caldwell, 1998; Choi, 2002), we still have significant gaps in fully understanding the team boundary-spanning phenomenon. For example, our knowledge remains incomplete regarding *how* teams differ in the pattern of boundary-spanning activities (e.g., do all members engage in boundary-spanning activities to

the same extent?) and *whether* this differentiation matters to teams' efforts in promoting creativity. In other words, although the existing literature has provided some appealing explanations regarding the crucial influences of team boundary spanning (e.g., Marrone et al., 2007; Somech & Khalaili, 2014), most of these claims are based on investigations of the mean level of boundary-spanning activities in teams and its impact on teams' task performance (Marrone, 2010).

Team members' differential involvement in boundary-spanning activities, however, is indeed common as it is usually determined by team structure and role specification. For example, in many teams, while some members take on internally directed tasks, other members are assigned with a networking role, taking responsibility for externally directed activities (Ancona, 1990). Also, individual differences in traits and characteristics that are relevant to boundary-spanning engagement (e.g., boundary-spanning self-efficacy; Marrone et al., 2007) may also affect individuals' willingness and commitment to engage in boundary-spanning activities. The exclusive focus on the aggregated, mean level either overly simplifies team boundary-spanning activities in the real workplace or erroneously assumes that all team members homogeneously engage in boundary-spanning activities to the same extent. Either way, it fails to capture the nuanced, subtle, and ongoing team processes, activities, and dynamics in teams of different configurations of boundary-spanning activity, such as how members of various levels of boundary-spanning engagement internalize information from outside, attend to competing internal and external demand and influence, and manage conflicting expectations, views, and ideas within teams (e.g., Friedman & Podolny, 1992; Marrone, 2010).

Moreover, although the bulk of past research has explored how team diversity affects team creativity (e.g., Shin & Zhou, 2007), we still know so little about the influences of diversity in members' boundary-spanning activities on team creativity. Past research has also noted that the impact of boundary activities on teams may not always be linear and positive (Choi, 2002), which also demonstrates the necessity of tapping into the configurations of boundary-spanning activity to better understand the team dynamics and functioning. Given that individual contributions may emerge at the team level in a non-isomorphic fashion (Kozlowski & Klein, 2000), teams with different configurations or patterns of boundary-spanning activities may incur unique team processes and internal dynamics, which consequently result in diverse outcomes (Klein & Kozlowski, 2000).

To advance our understanding about team boundary-spanning phenomenon and unpack its complex influences on team creativity, we focus on team member boundary-spanning differentiation (TMBSD) to exploratory probe the pattern/configuration of team boundary spanning. Our reading suggests that the categorization-elaboration model (CEM), proposed by Van Knippenberg et al. (2004), is a promising theoretical frame to answer our research questions regarding *whether*, *how*, and *when* TMBSD influences team creativity. On the one hand, the information/decision-making perspective proposes that team diversity can promote team functioning due to the inclusion of diverse knowledge and perspectives; the social categorization perspective, on the other hand, infers that diversity can impede

team functioning due to an “us versus them” mentality (Van Knippenberg et al., 2011). We believe that CEM provides a vantage point to elaborate the complex influences of team diversity in a comprehensive way, discussing both beneficial and adverse influences of team diversity by reconciling the seemingly competing arguments of these two perspectives.

Guided by CEM, we propose that the influence of TMBSD on team creativity can be non-linear if the magnitude of TMBSD significantly triggers the social categorization process, which limits the information elaboration process, through which TMBSD often promotes team creativity. As shown in Figure 1, we provide illustrative examples to visualize typical patterns/configurations of team boundary-spanning activities across teams of different levels of TMBSD. We argue that when moving from low to moderate level of TMBSD (see Figure 1a and b), TMBSD can be functional and promote team creativity, because the social categorization process is less likely to be activated and information can flow more smoothly within the team. However, when TMBSD further increases and reaches an overly high level (see Figure 1c, maximum TMBSD shows a clear split in roles and substantial separation diversity within the team), the adverse impact of social categorization processes becomes salient in teams and can severely limit internal information exchange and hamper team creativity.

We then develop a theoretical model to further explain *how* (i.e., information seeking) and *when* (i.e., high team goal interdependence) TMBSD will promote or inhibit team creativity. Specifically, we argue that information sharing, as a critical creativity-relevant team mechanism, serves as an essential conduit of transferring external information into teams, which makes it a critical process to realize the benefits of boundary spanning. Our investigation of the mediating role of information sharing heeds the call for research on information processing that takes place among members who possess divergent informational resource (e.g., Gilson et al., 2013). Moreover, according to CEM, the influence of team diversity on teams depends on the characteristics of the team's task and intragroup relations (Cooper et al., 2014). Interdependent team goals can increase communication and cooperation within a team through efficiently alleviating conflicts among team members caused by social categorization, and facilitating the integration and collective utilization of individual resources for pursuing the collective goal (Weldon & Weingart, 1993). Thus, we propose that the influence of TMBSD on team information sharing can be further moderated by team goal interdependence, which has been shown to be a critical task characteristic variable in understanding the influence of team boundary activities (Somech & Khalaili, 2014). Figure 2 shows the overall theoretical model.

Our research makes contributions to the team boundary spanning and creativity literatures in three ways. First, in contrast to the previous, sole emphasis on the mean/aggregated level of team members' behaviors and characteristics (e.g., Widmann & Mulder, 2018), this study extends the existing research by focusing on the pattern of teams' boundary spanning, with an important yet overlooked component of which being the differentiation within teams' boundary-spanning activities (i.e., TMBSD). Second, leveraging a CEM-based

FIGURE 1 Illustrative exemplars of different levels of team member boundary-spanning differentiation (TMBSD) and information sharing in teams at the same mean level of boundary-spanning activity. Note: All three teams in the exemplars have the same mean level (moderate) of boundary spanning. X-axis represents team members' participation in boundary-spanning activities. Members of various colors refer to their different level of engagement in boundary-spanning activities. $\langle \text{-----} \rangle$ and $\langle \text{=====} \rangle$ denote relatively weak and strong information exchange processes, respectively, as we hypothesized

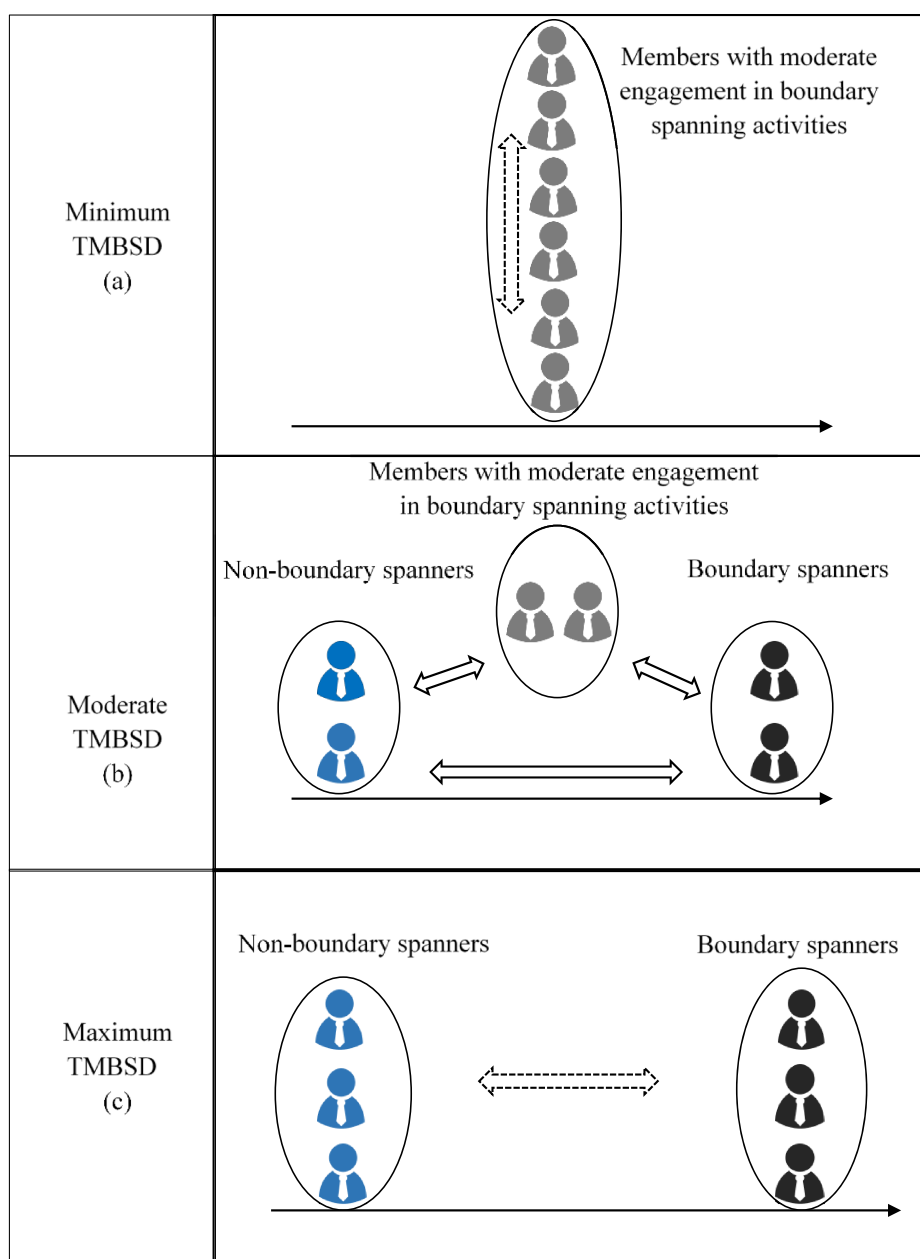
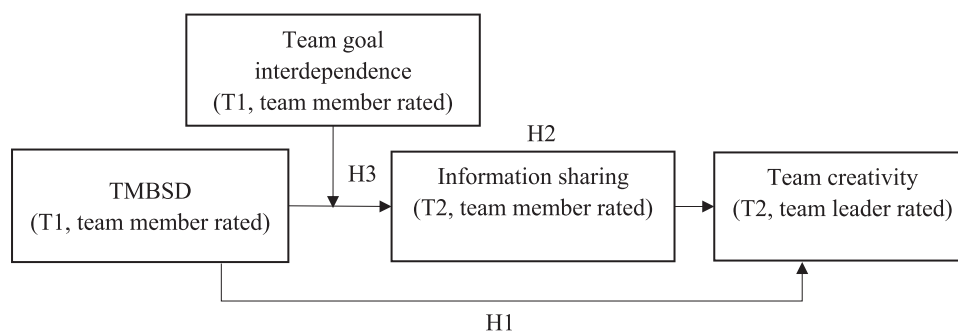


FIGURE 2 Hypothesized model. TMBSD, team member boundary-spanning differentiation



framework, we identify a complex, non-linear linkage between TMBSD and team creativity. In this way, we are able to unveil the range of TMBSD that activates more (less) information elaboration

process and less (more) social categorization process, both of which are proposed in CEM but affect team creativity in different ways. Our investigation of the underlying mechanism through information

sharing also provides an opportunity to gauge whether TMBSD influences team creativity as proposed by CEM. Finally, our consideration of team goal interdependence allows a comprehensive understanding of *when* TMBSD fosters/hampers team process and creativity. Our focus towards exploring the moderating role of team goal interdependence implies that not all teams passively react to the dynamics triggered by TMBSD, but rather appreciates that some can actively make efforts to ameliorate the potential disruptive impact of TMBSD and reap the benefits of TMBSD. Taken together, we respond to Marrone et al.'s (2007) call to address how teams can best design, utilize, and manage boundary-spanning activities to contribute to team outcomes.

2 | THEORY AND HYPOTHESES DEVELOPMENT

2.1 | Team member boundary-spanning differentiation

Team member boundary-spanning behaviors are defined as behaviors aimed to establish connections with external parties in order to help teams to achieve their objectives (Ancona & Caldwell, 1992; Marrone et al., 2007). Team boundary-spanning activities include various externally directed activities—some present an information-gathering function (e.g., boundary spanners try to search and transfer knowledge and information from the external environment), whereas others have an external representation and coordination function (e.g., boundary spanners build up connections with external stakeholders to adjust team objectives, request resources, update team progress, and coordinate task activities). Based on the social network perspective, boundary spanners carry out the responsibility of filling structural holes (Burt, 1992) so as to acquire task-related information and establish a stable relationship with the external environment. Thus, all dimensions of boundary-spanning activities can serve as critical means for teams to seek and maintain valuable sources of the problem- or project-specific information (e.g., feedback on projects, evaluation and new ideas of tasks, and technical and market information).

Boundary spanners usually embrace a significant, stable, and identifiable job role and identity (Marrone et al., 2007). For example, the marketers are mainly responsible for externally directed activities, whereas the engineers are mainly responsible for internally directed activities; this difference leads a marketer to embrace a typical boundary-spanner identity and leaves an engineer with a non-boundary-spanner identity. Research on identity construction supports our argument and clearly states that a role identity can develop and become internalized over time as a result of an individual retrospectively interpreting the continued role activities (Farmer et al., 2003). Once established, it is hard to change such identities in a short time. According to the social identity argument (Stets & Burke, 2000), a stable identity may indeed enable individuals to identify with similar members as the in-group, viewing the dissimilar others as the out-group. Research on subgroups in work teams also has proposed that

individuals tend to associate themselves with those they believe to share the same identity (Carton & Cummings, 2012).

Individual members vary in their engagement in boundary-spanning activities, partly because of the differences in job duties and partly because of their inclinations to engage in and commit to such behaviors (Marrone et al., 2007). Individuals' involvement in boundary spanning can vary along a continuous, lateral continuum and TMBSD reflects the overall pattern of such difference among team members. Different levels of TMBSD may result in different extents of split in teams. Again, as illustrated in Figure 1, the minimum level of split occurs in teams where all members engage in similar levels of boundary spanning (either at a very high or very low level), whereas the maximum level of split exists in teams when some members highly involve in boundary-spanning activities while others maintain a very low level involvement. Examples of moderate levels of split include teams where either all members uniformly differ from each other in their engagement in boundary-spanning activities or most members engage in similar levels of boundary spanning yet a few other members differ (Harrison & Klein, 2007).

In what follows, we invoke a CEM-based framework and begin by theorizing how TMBSD influences team creativity in a curvilinear fashion that is mediated by team information sharing. This study then investigates team goal interdependence as a critical contingency factor that moderates the influence of TMBSD on information sharing.

2.2 | TMBSD, information sharing, and team creativity

The division of roles regarding boundary-spanning activities creates both opportunities and challenges for teams. From the information/decision-making perspective, intrateam role diversity may increase members' exposure to diverse information and perspectives, which can strengthen the cooperative information elaboration within teams. More importantly, intrateam role differentiation not only reduces the possibility of information redundancy caused by collective participation, but also creates necessary gaps of knowledge and triggers epistemic motivation that further incentivizes members to engage in information exchange (Cheung et al., 2016). However, the self-categorization theory, as an important extension of social identity theory (Stets & Burke, 2000; Stryker & Statham, 1985; Turner et al., 1987), holds that the differences in roles, including the work roles that are central to the formation of individuals' overall identity or self-definition (Welbourne & Paterson, 2016), provide the basis for individual members to categorize self and the dissimilar others. Thus, this line of research often predicts that the heterogeneity in group roles is likely to cause harm (e.g., relational conflicts) to group functioning (Jehn et al., 1999).

To unveil the complex influences of the intrateam diversity regarding boundary-spanning roles (or TMBSD) on team creativity, we adopt the CEM to comprehensively elaborate the beneficial and harmful effects of TMBSD. As a promising framework to reconcile the seemingly competing arguments of the information

sharing perspective and the social categorization perspective, CEM offers a more balanced view that although diversity in role/identity can promote inclusion of diverse knowledge and perspectives, it may also impede team dynamics due to an “us versus them” mentality under certain circumstance (Van Knippenberg et al., 2004). Guided by CEM, we expect that the existence of TMBSD can be beneficial for teams at a low or moderate level, such as when most members engage in similar levels of boundary spanning yet a few other members differ. Within this range, the greater the role differentiation is, the stronger the members will perceive that others may possess important information to supplement their knowledge as well as the knowledge repository of the team. Members' recognition of the information gaps within the team is likely to motivate them to make more efforts to seek and share information and exchange novel ideas and creative viewpoints. Differentiation of boundary-spanning activities within the low to moderate range also enables the matching of members to their best-fitting roles/responsibilities (Li et al., 2015) and enhances the convenience of locating different task-specific information (e.g., members can directly ask boundary spanners for market information regarding market trends and technical evolution). In this way, the ease of parsing, routing, and sourcing task-related information based on member roles and responsibilities facilitates frequent and effective exchange of information within teams (Bunderson & Boumgarden, 2010). Moreover, according to the cross-categorization perspective (Chen et al., 2017), teams with cross-categorization members can reduce inter-subgroup communication bias and increase interpersonal liking, which helps teams to make use of every member's knowledge and give careful consideration to their diverse perspectives, thus promoting the collective information sharing in teams. In this regard, teams with moderate levels of TMBSD seem to be constructive in enhancing information sharing (see Figure 1b).

As the level of TMBSD continues to increase, however, we expect the social categorization process to play a greater role in team functioning by gradually splitting the team. At overly high levels of TMBSD (e.g., nearly half of team members act as boundary spanners while the rest of the team is involved in little boundary-spanning activities), teams may even form two polarized identities—boundary spanners and non-boundary spanners. According to the social categorization perspective and CEM, when teams are dominated by members with these two identities, identity conflicts are inevitable because different identities can activate in-group favoritism and out-group denigration associate with each identity (Turner et al., 1987). Thus, the resulting biases and stereotypes against others can severely restrict cooperative communication, damage cohesion, and hinder collaborative efforts on information sharing. In addition, the boundary spanners and non-boundary spanners may even develop different jargon and meanings for the same words (Ancona & Caldwell, 1989), which can lead to difficulties in interpretation, communication, and information exchange.

Once TMBSD increased to a certain point, the perceived large gap in identity would be insurmountable in the eyes of both the

boundary spanners and the non-boundary spanners, likely triggering hostility and harmful conflicts and frictions between members of different identities (Carton & Cummings, 2013). Such identity conflicts further increase cognitive load, which interferes with the exchange and processing of information (Jehn & Bezrukova, 2010) and distracts team members from developing new ideas or adopting new methods. This argument is also consistent with the previous study that the existence of multiple, especially opposing identities can be detrimental to interpersonal dynamics and team outcomes (Carton & Cummings, 2012). Once the perceptions regarding the salience of social categorization are consolidated, intrateam identity conflict gets further aggravated (Randel, 2002), and members of different identities tend to keep the valuable information private (Hutzschenreuter & Horstkotte, 2013). Thus, as TMBSD reaches an overly high level, we anticipate its adverse consequences to outweigh its benefits in promoting team creativity.

Our reasoning above suggests that information sharing serves as a critical mechanism in explaining the impact of TMBSD on creativity. Consistent with the boundary spanning and creativity literatures (Gong et al., 2013; Hahm, 2017), we believe that team information sharing not only helps members connect previously isolated ideas and knowledge, recombine existing ideas, and generate newness (Amabile, 1988; Gong et al., 2013), but also ignites creative sparks through facilitating the discussion of divergent viewpoints, prototype shortcomings, and innovative procedures (Van Knippenberg et al., 2011). In sum, compared to teams with a high level of TMBSD, teams with low or moderate levels of TMBSD are likely to reap the benefit of information sharing with little risk of engaging in the potential disruptive social categorization process, and therefore may ultimately promote team creativity. However, when TMBSD reaches an overly high level, the disruptive social categorization process is likely activated and information sharing will be accordingly impeded; consequently, team creativity decreases. We therefore hypothesize as follows:

Hypothesis 1. There is a curvilinear relationship between TMBSD and team creativity.

Hypothesis 2. Team information sharing will mediate the curvilinear relationship between TMBSD and team creativity.

2.3 | The moderating role of team goal interdependence

Team goal interdependence—the degree to which group members believe that they are assigned with group goals or provided with group feedback (Van der Vegt et al., 2003)—lies at the core of interpersonal relationships in groups (Halevy et al., 2011) and is highly relevant to team boundary activities and the implications (Somech & Khalaili, 2014). Guided by CEM, we posit that team goal

interdependence can act as a moderator in the curvilinear relationship between TMBSD and information sharing.

In teams with low to moderate levels of TMBSD, shared team goal directs members' attention and effort to cooperation and coordination for achieving collective performance rather than individual performance (Van der Vegt et al., 2001). With the recognition that others may possess diverse information and alternative channels for information, such teams will be more motivated to interact and communicate to facilitate the completion of their shared goals (Chen et al., 2011). Therefore, at moderate levels of TMBSD, goal interdependence can facilitate the information flow and exchange within teams, particularly between boundary spanners and non-boundary spanners, thus promoting information sharing within teams. In contrast, in teams with low to moderate levels of TMBSD but low goal interdependence, members are fully aware of their differences in roles they assume and the individual goals they desire to achieve, and thus may view coordination within teams, including that among boundary spanners and non-boundary spanners, as unnecessary in promoting individual success (Chen et al., 2008), which leads to lower information exchange among members.

In teams with overly high TMBSD, as we discussed earlier, the likelihood of identity and interpersonal conflict continues to increase, which prohibits the process of information seeking and exchange within teams. However, we expect that high goal interdependence may buffer these disruptive influences of TMBSD, given the evidence that goal interdependence can reduce interpersonal frictions and transform team members' self-perceptions from "us versus them" to the more inclusive "we" (Somech & Khalaili, 2014) and encourage the formation of high-quality social exchange relationships to integrate the knowledge of interdependent members in meeting task demands (Barrick et al., 2007). With goal interdependence, members are more likely to rationalize the differences in role arrangement as well as TMBSD, which in turn mitigates the risk of the disruptive social categorization mechanism, advocates collective orchestration of resources for pursuing superordinate goals (Weldon & Weingart, 1993), and further fosters the cooperation and interaction among boundary spanners and non-boundary spanners. In contrast, in teams with high TMBSD but lower goal interdependence, cooperation is perceived to be less critical, and thus further widens the gulf between members with different roles and identities.

Taken together, we expect that when team goal interdependence is high rather than low, the positive relationship between TMBSD and information sharing at low to moderate levels of TMBSD will be enhanced and the negative relationship between TMBSD and information sharing at high levels of TMBSD will be weakened. These predictions lead to the following hypothesis:

Hypothesis 3. Team goal interdependence moderates the curvilinear relationship between TMBSD and information sharing.

3 | METHOD

3.1 | Sample and procedure

Our data were collected from 70 R&D teams in 18 companies from Northern China. These firms are from industries of telecommunications, materials and equipment manufacturing, software development and system integration, biotechnology and pharmaceuticals, and so forth. The surveyed teams are particularly suited for this study, because all teams require some levels of boundary-spanning activities with external teams or organizations and it is common for members to vary in their engagement of boundary-spanning activities. Job positions of surveyed members include software engineers, new product developers, market researchers/analysts, and/or business marketers, and so forth. Also, all these teams deal with knowledge-intensive tasks that require the development of new products and/or services.

We collected our data from two different sources (team members and their corresponding team leaders) at two time points across 4 months to minimize potential common method bias. At Time 1, team members reported individual characteristics, boundary-spanning behaviors, and team goal interdependence. Four months later (i.e., at Time 2), team members rated team information sharing and team leaders evaluated teams' creativity. The survey process was anonymous to guarantee the privacy of participants.

A total of 398 members in 82 teams participated in our survey. Previous research has shown that calculating diversity indices with incomplete attribute data can distort the true diversity–outcome relationships (Allen et al., 2007). Therefore, teams with very low response rates were excluded from our data analysis. Following Dawson (2003) and Richter et al.'s (2006) guidance, the cut-off point was a selection rate ($([N - n]/Nn)$ of 0.32 (n = the number of responses per team and N = team size). Teams with selection scores of 0.32 or below can be included in our final sample. After excluding the teams did not meet this cut-off point, the resulting sample consisted of 315 team members from 70 teams. The response rate was 85.4% among teams and 79.1% among members. We also tested the similarity between the responded and non-responded individuals and teams to check for any possible non-response bias. The results of t tests show that the responded and non-responded individuals did not significantly differ in age and educational level, and the responded and non-responded teams did not significantly differ in team tenure and team size. Therefore, the non-response bias was not a serious concern in this study.

The average team size in this study was 4.5. Among all the team leaders, male leaders account for 75.7% of them, the average age of them was 36.94 ($SD = 5.60$), 96.7% of them had obtained a bachelor's degree or higher, and the mean team tenure was 10.54 years ($SD = 7.82$). Among team members, males accounted for 57.8%, the average age was 30.55 ($SD = 5.90$), 90.5% had obtained a bachelor's degree or above, and the mean team tenure was 4.86 years ($SD = 5.30$).

3.2 | Measures

All variables were rated on a 7-point Likert-type scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). The research team translated all the scales from English into Chinese. And then, standard back-translation procedures were followed to make sure of the equivalent reflection (Brislin, 1980). We discussed and resolved any inconsistencies to ensure the clarity of the survey instruments. (Please see Appendix A for the full-scale items of the core variables.)

3.3 | Team member boundary-spanning differentiation

Following existing research, the 6-item scale adapted from Marrone et al.'s (2007) study was used to measure individual boundary-spanning behavior. A sample item is "I reach out to individuals outside of our team that can provide project-related expertise or ideas." (Cronbach's $\alpha = .81$) Following guidelines for assessing separation diversity (Harrison & Klein, 2007), we calculated TMBS using the within-group standard deviation of individual boundary spanning.

3.4 | Information sharing

We used the 4-item scale in Bunderson and Sutcliffe's (2002) study to measure information sharing. A sample item is "Information used to make key decisions was freely shared among the members of the team." (Cronbach's $\alpha = .89$).

3.5 | Team goal interdependence

A 6-item scale was adopted to assess team goal interdependence (Van der Vegt et al., 2001). A sample item is "We are informed about the goals we should attain as a group." (Cronbach's $\alpha = .90$).

3.6 | Team creativity

We followed previous research (Gong et al., 2013) and invited team leaders to rate the team creativity of their own team with a 4-item scale developed by Shin and Zhou (2007). A sample item is "How well does your team produce new ideas?" (Cronbach's $\alpha = .85$).

3.7 | Control variables

We controlled team size, team tenure, and several diversity factors (i.e., team tenure diversity, gender diversity, education diversity, and functional diversity), as these variables were shown to influence social categorization processes and team creativity (Harrison et al., 2002). We calculated the coefficient of variation of team members' team

tenure (standard deviation divided by the mean) to represent team tenure diversity (Koopmann et al., 2014). We used Blau's (1977) index to measure education diversity, gender diversity, and functional diversity. In addition, we calculated the within-group mean of individual-level boundary spanning and controlled it in our analysis, because previous research has justified it as an important indicator of team innovation (e.g., Widmann & Mulder, 2018).

4 | RESULTS

We calculated three statistical indices to justify data aggregation to team levels. First, we computed the within-team inter-rater agreement (R_{wg}). For information sharing, the average R_{wg} was .88, and for team goal interdependence, the average R_{wg} was .95, both of which support aggregation of the data (Bliese, 2000). Second, intra-class correlation (ICC_1) was .25 for information sharing and .28 for team goal interdependence. Third, the reliability of the mean (ICC_2) was .61 for information sharing and .63 for team goal interdependence. All these indices indicate that the aggregation is justified. We also conducted confirmatory factor analysis (CFA) to examine the distinctiveness of all the variables, including team boundary spanning, information sharing, team goal interdependence, and team creativity. The results presented that the four-factor model showed a better overall fit ($\chi^2/df = 1.16$, CFI = .97, TLI = .97, RMSEA = .05) than other models with fewer factors.

4.1 | Hypothesis testing

Table 1 presents the descriptive statistics, correlations, and scale reliabilities for all the variables.

All the results of hierarchical regression analyses to test Hypotheses 1–3 are shown in Table 2. We included all the control variables and team member boundary-spanning mean in all models.

Hypothesis 1 proposes a curvilinear relation between TMBS and team creativity. In Model 2 of Table 2, after entering TMBS and its quadratic term, the coefficient of the linear term of TMBS was not significant ($\beta = .01$, *n.s.*), whereas the quadratic term was significantly and negatively associated with team creativity ($\beta = -.45$, $p < .01$), which indicates that TMBS is curvilinearly related to team creativity. Following the guidelines of Aiken and West's study (1991), we depicted a graphical representation of the above relationship in Figure 3. At lower levels of TMBS, the relationship between TMBS and creativity showed a positive trend, whereas at higher levels of TMBS, the relationship between TMBS and creativity becomes negative, supporting Hypothesis 1.

To justify Hypothesis 2, Model 6 of Table 2 included both TMBS and its quadratic term. The results suggested that the quadratic TMBS showed a significant and negative influence on team information sharing ($\beta = -.50$, $p < .01$). We depicted this curvilinear relationship in Figure 4, which showed that information sharing increased when TMBS moved from the low level to the moderate

TABLE 1 Means, standard deviations, and intercorrelations of variables

Variable	M	SD	1	2	3	4	5	6	7	8	9	10	11
1. Team tenure	4.85	3.79											
2. Team size	4.50	1.35	−0.02										
3. Gender diversity	0.34	0.18	0.08	0.25*									
4. Tenure diversity	0.66	0.30	0.06	−0.12	−0.16								
5. Education diversity	0.27	0.24	−0.09	0.12	−0.05	0.04							
6. Functional diversity	0.31	0.22	0.10	0.09	0.23	0.20	0.18						
7. Team member boundary-spanning behavior mean	5.37	0.40	−0.05	−0.07	−0.11	−0.05	−0.05	0.23	(0.73)				
8. Team member boundary-spanning differentiation (TMBSD)	0.64	0.34	−0.07	0.17	0.05	0.09	0.09	−0.04	−0.55**				
9. Team information sharing	5.49	0.64	−0.24*	0.07	0.05	−0.16	0.14	−0.15	0.20	−0.13	(0.91)		
10. Team goal interdependence	5.36	0.50	0.08	0.06	0.10	−0.05	−0.04	−0.01	0.45**	−0.18	0.12	(0.83)	
11. Team creativity	5.60	0.77	−0.08	0.11	0.02	−0.06	0.12	−0.14	−0.04	0.00	0.52**	0.05	(0.83)

Note: $n = 70$; the figures on the diagonal in parentheses are the alpha coefficients.

* $p < .05$. ** $p < .01$.

TABLE 2 Results of hierarchical regression analysis

	Team creativity				Information sharing		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
Team tenure	−0.05	−0.04	0.06	0.04	−0.20	−0.18	−0.23*
Team size	0.11	0.04	0.08	0.05	0.07	−0.03	−0.11
Gender diversity	−0.05	0.02	−0.05	−0.02	0.01	0.08	0.15
Tenure diversity	−0.02	0.08	0.04	0.08	−0.12	−0.01	−0.02
Education diversity	0.13	0.09	0.04	0.04	0.16	0.13	0.11
Functional diversity	−0.17	−0.10	−0.06	−0.05	−0.20	−0.11	−0.05
Team member boundary-spanning behavior mean	−0.01	−0.09	−0.14	−0.17	0.24*	0.17	0.09
TMBSD		−0.01		−0.03		0.07	0.05
TMBSD squared		−0.45**		−0.23		−0.50**	−0.36**
Team information sharing			0.56**	0.44**			
Team goal interdependence							0.25
TMBSD $\times$ team goal interdependence							0.37**
TMBSD squared $\times$ team goal interdependence							−0.16
R^2	0.05	0.23	0.31	0.35	0.17	0.37	0.45
Change in R^2		0.17**	0.25**	0.12**		0.20**	0.08*

Note: $n = 70$; standardized regression coefficients are reported.

* $p < .05$. ** $p < .01$.

level, whereas, once TMBSD moved away from a certain level, team information sharing peaked and started to decline. Besides, team information sharing had a positive effect on team creativity ($\beta = .56$, $p < .01$; see Model 3), and the coefficient of information sharing remained significant ($\beta = .44$, $p < .01$; see Model 4) even after we included TMBSD and its quadratic term into the model. Taken

together, these results suggest that TMBSD is curvilinearly related to information sharing, and team information sharing mediates the effects of TMBSD on team creativity, thus supporting Hypothesis 2.

We then adopted the bootstrapping analysis and used 5000 bootstrap samples to further test Hypothesis 2. Using the recommended mediation approach by Hayes and Preacher (2010), we examined the

FIGURE 3 The relationship between team member boundary-spanning differentiation (TMBSD) and team creativity. High TMBSD, high team member boundary-spanning differentiation; low TMBSD, low team member boundary-spanning differentiation

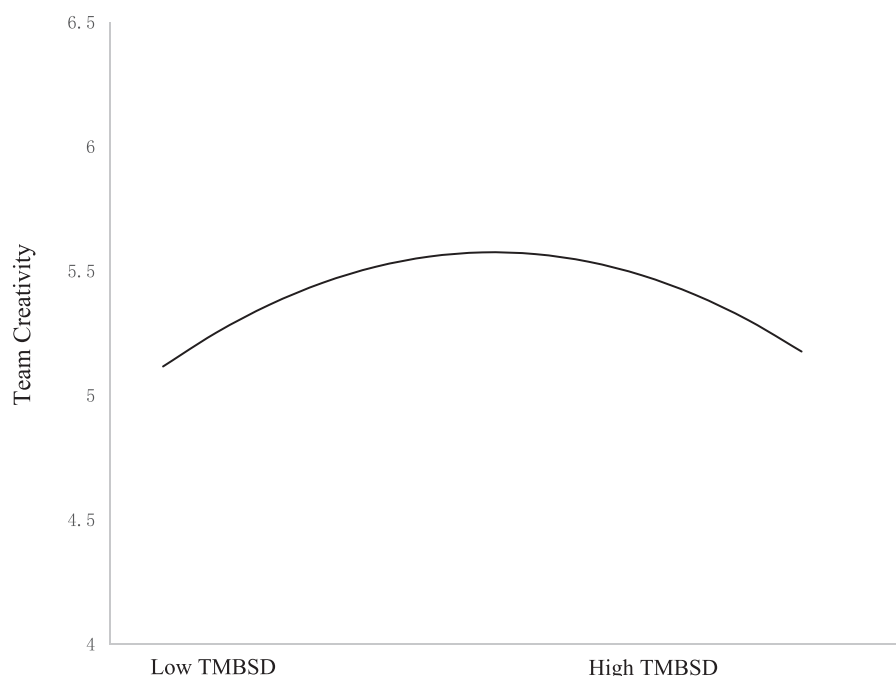
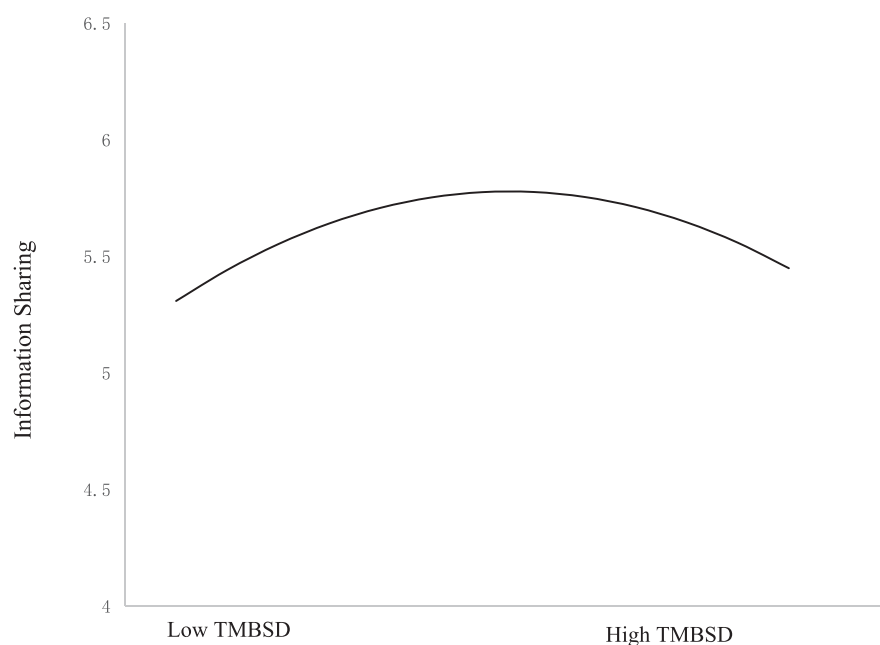


FIGURE 4 The relationship between team member boundary-spanning differentiation (TMBSD) and team information sharing. High TMBSD, high team member boundary-spanning differentiation; low TMBSD, low team member boundary-spanning differentiation



indirect effect of TMBSD on creativity via information sharing at lower and higher degrees of TMBSD (i.e., -1 SD, $+1$ SD). The results show that at both the lower (.52, bias corrected bootstrap CI = 95% [.08, 1.19]) and higher levels of TMBSD ($-.38$, bias corrected bootstrap CI = 95% [$-.79$, $-.05$]), the indirect effect was significant, indicating that an increase of TMBSD in teams with low levels of TMBSD could lead to higher creativity through an increase in information sharing; on the contrary, an increase of TMBSD in teams with high levels of TMBSD would result in lower creativity through a decrease in information sharing. These results again support Hypothesis 2.

We included team goal interdependence into the regression model to test Hypothesis 3. In Table 2, the results of Model 7 show that TMBSD interacted with team goal interdependence to significantly influence team information sharing ($\beta = .37$, $p < .01$), when the TMBSD squared term was also inserted in the model. The significance of this interaction indicates that the slope of the relationship between TMBSD and information sharing was a function of team goal interdependence. However, the interaction between team goal interdependence and the TMBSD squared term was not significant ($\beta = -.16$, *n.s.*), suggesting the curvilinear relationship was constant at

different degrees of team goal interdependence. Overall, the shapes of the TMBSD–team creativity relationship differ significantly between relatively lower and higher levels of team goal interdependence. Therefore, Hypothesis 3 is supported. Furthermore, we graphed Figure 5 (Aiken & West, 1991) to illustrate the curves between TMBSD and team information sharing at high and low levels of team goal interdependence (i.e., $+1$ SD/ -1 SD). Figure 5 showed that when TMBSD is within the low to moderate range, high rather than low team goal interdependence will strengthen the positive relationship between TMBSD and team information sharing; when TMBSD gets higher than the moderate level, high rather than low team goal interdependence will weaken the negative relationship between TMBSD and team information sharing.

4.2 | Supplementary analyses

We ran supplementary analyses to alleviate concerns of alternative mechanisms and provide further support to our theoretical rationales. To investigate whether members may indeed form different identities¹ according to their participation in boundary-spanning activities as we proposed, we conducted a supplementary analysis testing team member reported boundary-spanning behaviors (Time 1), their perceptions of boundary spanner identity at 4 months later (Time 2), and the relationship between their behavior and identity. We followed Farmer et al.'s (2003) approach to measure the extent to which the role of boundary spanners had been incorporated into self-identity. The items were modified to reflect the centrality of role identity as a boundary spanner, a sample item is “I have clear concept of myself as a boundary spanner” ($\alpha = .90$). We observed substantial variance in individual self-reported boundary spanner identity ($M = 3.96$, $SD = 1.36$) across teams, indicating that TMBSD has indeed occurred in our sample. Our data show that team member boundary-spanning behavior was significantly and positively related to an individual's

boundary spanner identity ($\beta = .27$, $p < .01$), suggesting that the boundary-spanner identity could have formed after their participation into boundary-spanning activities.

Given that the conflict between different identity-based subgroups (boundary spanners vs. non-boundary spanners) is another key premise underlying our theoretical rationale, we measured team conflict in Time 2 with a 6-item scale in Bendersky and Hays's (2012) study. Regression results show that TMBSD was positively related to team conflict ($\beta = .25$, $p < .01$), supporting our arguments that high TMBSD can indeed create conflicts within teams. These supplementary analyses strengthened the theoretical arguments underlying our hypotheses.

5 | DISCUSSION

Although considerable efforts have been made to better understand the team boundary spanning and its impact on teams, past work has been preoccupied with answering the question: “Whether more boundary-spanning activities lead to better team outcomes?” The pattern of team member boundary-spanning activities, particularly the differentiation across members has been overlooked, and its influence on an important team outcome, team creativity, remains largely unknown. We believe that the time is ripe to further decipher team boundary spanning, particularly its intrateam differentiation (TMBSD), and its impact on team creativity, partly by answering advanced questions such as, “Whether too little or too much of TMBSD fosters creativity” and “When and how TMBSD fosters creativity?” Answering these questions allow for a more accurate understanding of teams' internal dynamics regarding boundary spanning, especially those teams where member differ substantially in their role assignments and engagement in boundary-spanning activities.

To explicate the process linking TMBSD to team creativity, we use CEM as an overarching theoretical framework to argue that

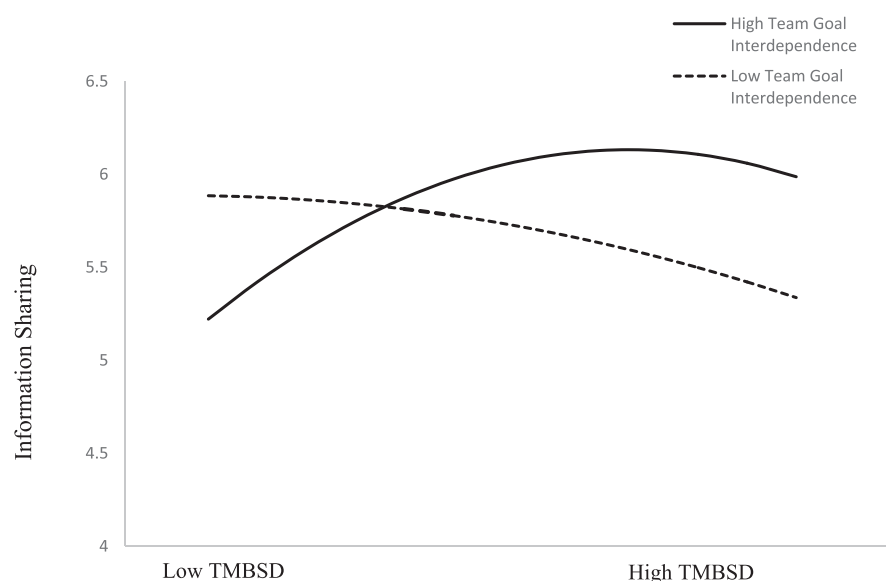


FIGURE 5 The moderating effect of team goal interdependence on the team member boundary-spanning differentiation (TMBSD)—information sharing relationship. High TMBSD, high team member boundary-spanning differentiation; low TMBSD, low team member boundary-spanning differentiation

although TMBSD may provide benefits for teams, levels of TMBSD can indeed lead to subgroup creation and identity conflicts that restrict communication and undermine information sharing within teams. Consistent with our theorization, we found that the TMBSD—team creativity relationship followed a curvilinear shaped relation and was mediated by information sharing. Moreover, we verified that the curvilinear relation between TMBSD and information sharing was moderated by team goal interdependence, such that the shapes of the TMBSD—team creativity relationship differ significantly between relatively lower and higher levels of team goal interdependence. Specifically, our results suggest that moving from low to moderate levels of TMBSD, the positive effect of TMBSD on team creativity was enhanced in teams with higher rather than lower goal interdependence; and the negative impact of TMBSD on team creativity at high levels of TMBSD was weakened in teams with higher rather than lower goal interdependence.

Although our hypothesized model has received general support using the data we collected at two time points from multiple sources, the findings we observed in Figure 5 suggest areas that remain unexplained by our hypotheses. For example, at overly low levels of TMBSD, we found that teams with higher goal interdependence demonstrated an even lower level of information sharing than teams with lower goal interdependence. We suspect shared team goal may mitigate the process of information sharing among members in teams of low TMBSD for two reasons. First, if these members shared the belief that they were assigned with the same group goals and/or received identical group feedback (i.e., high goal interdependence), they would likely to see each other to be on the “same page” and would consider additional information seeking within the team as unnecessary as it brings little additional benefit and may even result in information overlap and redundancy. Thus, they are less likely to seek information from others whom they assume to be alike, and more likely to side-step confrontation and avoid the risk of “rocking the boat” by consciously limiting the introduction of new facts or the sharing of diverse views. Second, in teams with overly low TMBSD, shared goals may also lead members, especially those who perceive themselves to be highly alike in roles and responsibilities to favor a default plan or certain solutions before group discussion, which imposes additional risks of groupthink² (Devine, 1999). They may view harmony and conformity in group results as an effective strategy to achieve their shared goal, which may suppress information sharing and promote compromised, mediocre solutions to teams (Aldag & Fuller, 1993; Ellis et al., 2003).

5.1 | Theoretical contributions

This study is one of the first studies to offer a theoretical conceptualization and operationalization of the relationship between TMBSD and team creativity. Our investigation of the mediating mechanism and the boundary condition adds another important layer of understanding to the complex effects of team boundary spanning. Our research makes three contributions to the literature. First, we extend

boundary-spanning theory by making a pioneering investigation on the effect of differentiation of boundary spanning within the team, TMBSD, on team creativity. It is in contrast to the previous, sole emphasis on the mean/aggregated level of team members' behaviors and characteristics (e.g., Widmann & Mulder, 2018). Prior study offered compelling evidence of how mean-level boundary-spanning influences team internal process and outcomes; however, it fails to specify how teams carry out these activities and the resulting impacts of the non-isomorphic individual behavioral contributions to team boundary spanning (Marrone, 2010). Our theoretical attention to variability in individual activities could complement the previous evidence, thus allowing for a more nuanced view of teams' boundary-spanning activity and its resulting outcomes. In this way, this study extends the research on understanding the antecedents of team creativity from focusing on the impacts of aggregated, mean level of team behavior to investigating the different configurations and patterns of team behavior.

Second, leveraging a CEM framework, we theoretically explain how TMBSD will have a non-linear influence on team creativity. In doing so, we are able to unveil the range of TMBSD that activates more (less) information elaboration process and less (more) social categorization process, both of which substantially affect team creativity but in the opposite ways. Our results suggest that moderate TMBSD can be most beneficial to team creativity due to the low potentiality of the activation of the social categorization process and the benefits of information sharing. However, when TMBSD goes beyond a certain level, the gradually activated social categorization process will impede information sharing among members and ultimately inhibit team creativity. Through revealing how teams with high TMBSD would split into identities of boundary spanners and non-boundary spanners (as suggested in supplementary analysis), we also provide supportive evidence to Sui et al.'s (2016) findings that maximum separation is likely to activate the formation of two extreme and opposing subgroups, which generates intragroup conflicts and undermines team coordination and communication. In addition, our investigation of the underlying mechanism through information sharing not only provides another opportunity to gauge whether TMBSD influences team creativity in the way as proposed by CEM but also supports the existing research findings of the positive associations of information diversity and sharing with team creativity (e.g., Amabile, 1988; Hahm, 2017).

Finally, our efforts to examine team goal interdependence as a moderator offer important insights into the contingency factors of the effect of TMBSD on internal team functioning, and thus answering the important question of “when” TMBSD functions as expected. Our focus on the moderating role of team goal interdependence also reflects the view that not all teams passively react to the dynamics triggered by TMBSD, but rather appreciates that some can actively make efforts to ameliorate the potential disruptive impact of TMBSD and better reap the benefits of TMBSD. Together, combined with the investigation of the curvilinear nature of the TMBSD—team creativity relationship and the identification of the information sharing as a mediator linking, this study was able to expand our understanding of *whether*, *when*, and *how* TMBSD affects team creativity.

5.2 | Managerial implications

Our study also offers some important implications for practitioners. First, the curvilinear effect of TMBSD on creativity promotion indicates that, despite the challenges, it is important for teams to acknowledge the importance of assigning boundary-spanning roles to some members and strive to seek the “optimal” level of TMBSD that is considered to be most beneficial to team creativity. Accordingly, team leaders or managers should pay attention to monitor (via interviews or surveys) members' levels of interest and actual engagement in boundary-spanning activities as well as their perceived differentiation in roles regarding boundary-spanning activities, and frequently seek for feedback from all members to recraft and rearrange the boundary-spanning roles. If TMBSD reaches a too high level, which leads to undesirable attitudes and morale in teams, leaders should conduct timely adjustments to avoid the formation of unnecessary subgroups, reduce potential conflicts, and promote team communication. Meanwhile, team leaders should also be aware if TMBSD may systematically overlap with team structure or demographic diversity (e.g., gender and tenure), which may further consolidate TMBSD and create challenges to adjust it. Also, if TMBSD reaches a too low level or all members assume identical roles, leaders should be mindful of the risks of groupthink that may further impede information sharing, especially in groups where members share a common group goal.

Second, our result informs team leaders and managers that they should continue to encourage information sharing within teams especially when facing extreme levels of TMBSD. Team leaders can encourage informal and formal events (e.g., hold regular sharing meetings twice a week) to promote team members' information-sharing behavior as well as the open, constructive discussion of conflicts; leaders may also reward information sharing to incentivize knowledge integration and ultimately the generation of novel ideas (Men et al., 2017).

Third, given our result that the combination of low TMBSD and high goal interdependence may indeed harm information sharing processes, we recommend team leaders to hold a balanced view when promoting team goal interdependence. An implication of our results is that higher team goal interdependence generally benefits teams with moderate and high levels of TMBSD and thus, in these teams, managers may consider devoting efforts to reinforce common goals, designing a shared reward distribution system, and cultivating cooperative norms and cultures to encourage team members to perform their tasks interdependently. However, in teams with low TMBSD, team leaders may consider setting both collective goals and individual goals to avoid the potential risks of information withholding and groupthink.

5.3 | Limitations and future research directions

Several limitations of our research provide possible directions for future study. First, although we collect our data from two different sources at two different times, future studies could design better studies or take advantage of experiment settings to further facilitate causal inference. Second, future study can build upon the results of

our study to further explore the possible, non-linear impact of TMBSD on other team outcomes, such as team effectiveness and team performance. Third, this study only examines the mediating role of information sharing through which TMBSD facilitates or impedes team creativity; however, research is needed to examine directly the other mechanisms that have been mentioned in this study (e.g., groupthink). This way, alternative mechanisms guided by other theoretical frameworks beyond CEM could also be explored. Fourth, alternative contextual factors may also influence the focal relationship of interest, such as task uncertainty, team leadership style, and team composition, all of which deserve in-depth investigation.

As one of the first endeavors to understand the role of TMBSD on team creativity, this paper starts with an operationalization of the within-group standard deviation of individual boundary spanning to test TMBSD; however, we encourage future studies to develop more nuanced measures and better designs to replicate the findings of our study. Thus, alternative ways³ of measuring patterns of boundary-spanning activities could provide additional information to help us fully capture the typology and conceptualization of TMBSD (e.g., Harrison & Klein, 2007; Sui et al., 2016) and further clarify the configurations of boundary spanners and/or non-boundary spanners across teams. For example, a team member may switch the role between boundary spanners and non-boundary spanners on different tasks and at different time points, and such experience of dual roles may also help reduce the stereotypes and biases against any subgroups, creating opportunities for interpersonal exchanges and information sharing among different members. Also, we acknowledge that measuring whether and how much team members perceive differences in other members' boundary-spanning activities or the formation of subgroups based on boundary spanner roles (e.g., Jehn & Bezrukova, 2010) helps reflect the core meanings of TMBSD. In our study, we did not include a repeated measure to capture possible changes in members' boundary-spanning activities; however, the dynamics in TMBSD is a promising research topic and may offer us more insights on how teams can best manage the boundary-spanning activities.

Finally, we conducted our research with a sample from China, and thus, the collectivistic culture may weaken the impacts of TMBSD on team creativity as well as the moderating role of group goal interdependence, because individuals may strive to actively make contributions to their teams as they strongly believe that the needs and goals of the group as a whole always outweigh the needs and desires of themselves. However, given that our data still support our hypotheses, we expect the proposed relationships to be more salient in other cultures of relatively higher individualism and lower collectivism. Therefore, it would be worthwhile to replicate the investigation of this study in different cultural contexts.

6 | CONCLUSION

Based on the CEM, we proposed and empirically tested *whether*, *how*, and *when* TMBSD will influence team creativity. The data supported

our hypotheses that there is a curvilinear relationship between TMBSD and team creativity that is transmitted by team information sharing. Also, we showed that team goal interdependence moderated the curvilinear TMBSD–team information sharing relationship. Taken together, we hope our study provides useful advancement in team boundary-spanning and creativity literatures.

ACKNOWLEDGMENT

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

ENDNOTES

- ¹ We thank an anonymous reviewer for this point.
- ² We thank an anonymous reviewer for this insightful point.
- ³ We are grateful for this insightful comment from an anonymous reviewer.

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APPENDIX A: KEY MEASURES

Team member boundary-spanning behavior (Marrone et al., 2007)

1. I persuade outsiders to support team decisions.
2. I prevent the outsiders from “overloading” the team with too many requests.
3. I reach out to individuals outside of our team that can provide project-related expertise or ideas.
4. I proactively seek the advice and support of our supervisor.
5. I scan the environment inside or outside the organization for technical ideas/expertise.
6. I scan the environment inside or outside the organization for marketing ideas/expertise.

Information sharing (Bunderson & Sutcliffe, 2002)

1. Information used to make key decisions is freely shared among the members of the team.
2. Team members work hard to keep one another up to date on their activities.
3. Team members are kept “in the loop” about key issues affecting the business unit.

Team goal interdependence (Van der Vegt et al., 2001)

1. We receive feedback about our team performance.
2. We are collectively held accountable for our team performance.
3. We receive regular feedback about our team functioning.
4. We are informed about the goals we should attain as a group.
5. We regularly receive information about what is expected from our team.
6. We have several clear targets we have to attain as a group.

Team creativity (Shin & Zhou, 2007)

1. How well does your team produce new ideas?
2. How useful are those ideas?
3. How creative do you consider your team to be?
4. How significant are those ideas to your organization?